

Data Converters

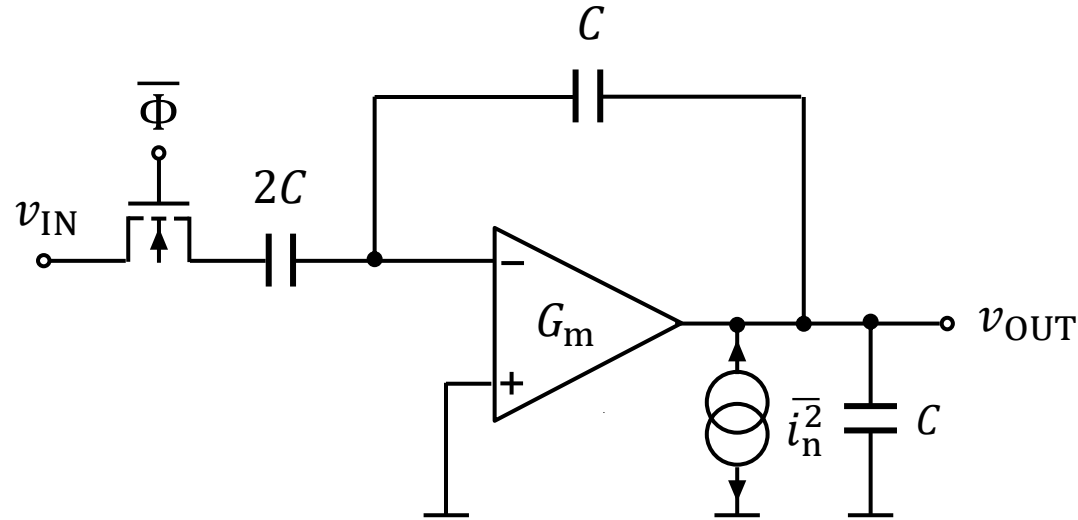
Exercise VI

Summer Term 2026

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TI.a – Output Noise Power G_m -Cell

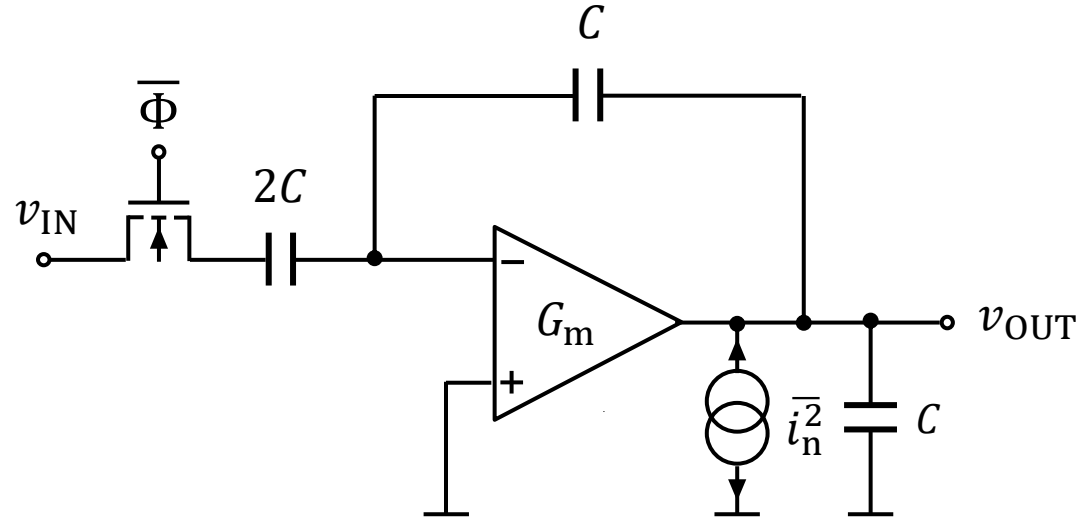


In order to find the output voltage noise power due to $\overline{i_n^2}$, we have to multiply the equivalent voltage noise power spectral density $\overline{v_n^2}$ by the equivalent noise bandwidth NBW .

$$\overline{v_{n,out,G_m}^2} = NBW \cdot r_{out_F}^2 \cdot \overline{i_n^2} = \frac{1}{4} \frac{\beta G_m}{C_{L_{tot}}} \cdot \frac{1}{\beta^2 G_m^2} \cdot n_f 4kT \gamma G_m = \frac{n_f \gamma}{\beta} \frac{kT}{C_{L_{tot}}}$$

TI.a – Output Noise Power G_m -Cell

$$\beta = \frac{C_F}{C_F + C_S} = \frac{1}{3}$$



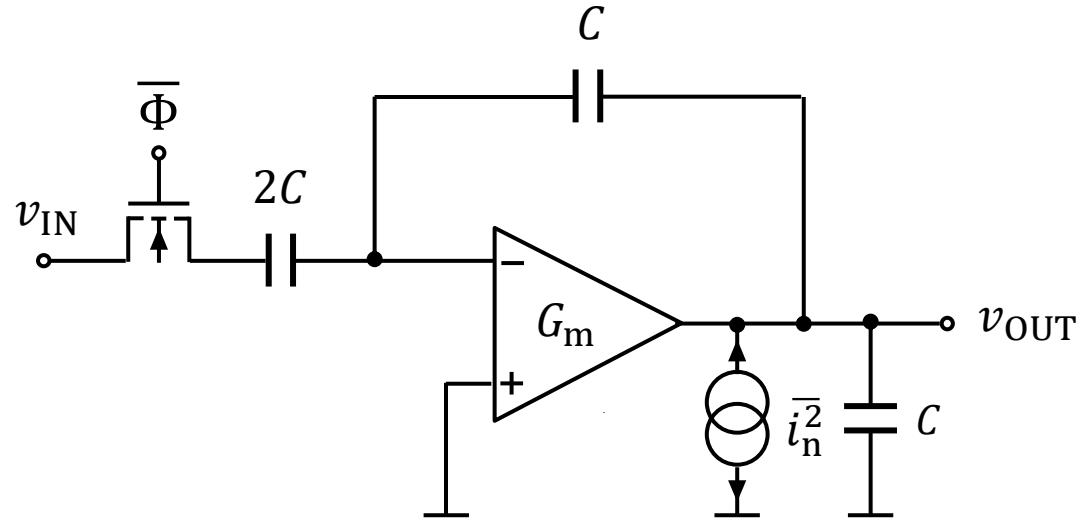
$$C_{L_{tot}} = C_L + (1 - \beta)C_F$$

$$= \frac{5}{3}C$$

$$\overline{v_{n,out,G_m}^2} = \frac{n_f \gamma}{\beta} \frac{kT}{C_{L_{tot}}} = n_f \gamma \frac{kT}{5/9 C} = 2 \frac{2}{3} \frac{1.38 \cdot 10^{-23} \text{ J/K} \cdot 293 \text{ K}}{5/9 \text{ pF}} \approx 9704 \mu\text{V}^2$$

$$v_{n,out,G_m,RMS} = \sqrt{\overline{v_{n,out,G_m}^2}} \approx 99 \mu\text{V}$$

TI.b – Output Noise Power Switch

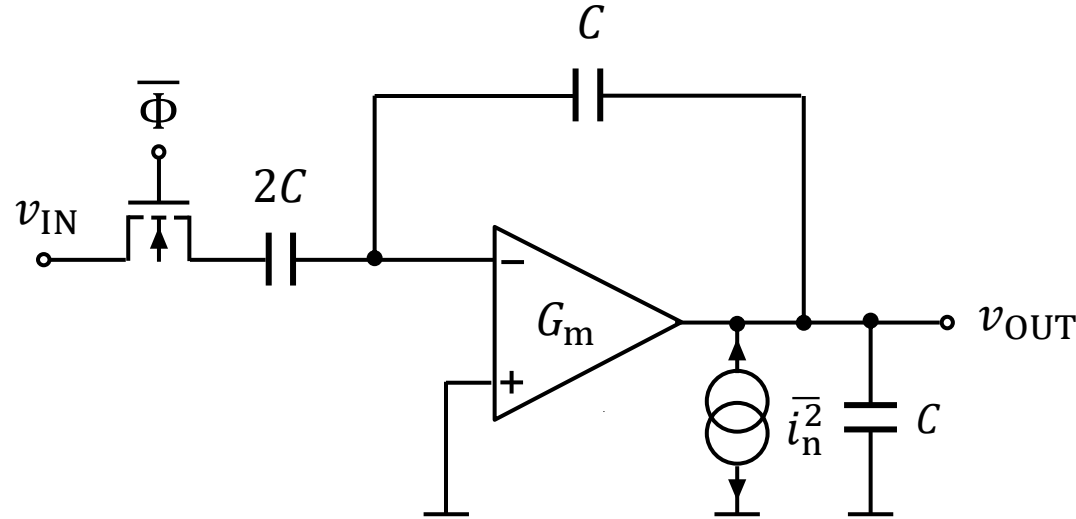


The power spectral density of the thermal noise of the switch will be transferred by the squared magnitude of the signal transfer function to the output.

There, it is to be multiplied by the equivalent noise bandwidth NBW in order to get the noise power.

TI.b – Output Noise Power Switch

$$\beta = \frac{C_F}{C_F + C_S} = \frac{1}{3}$$

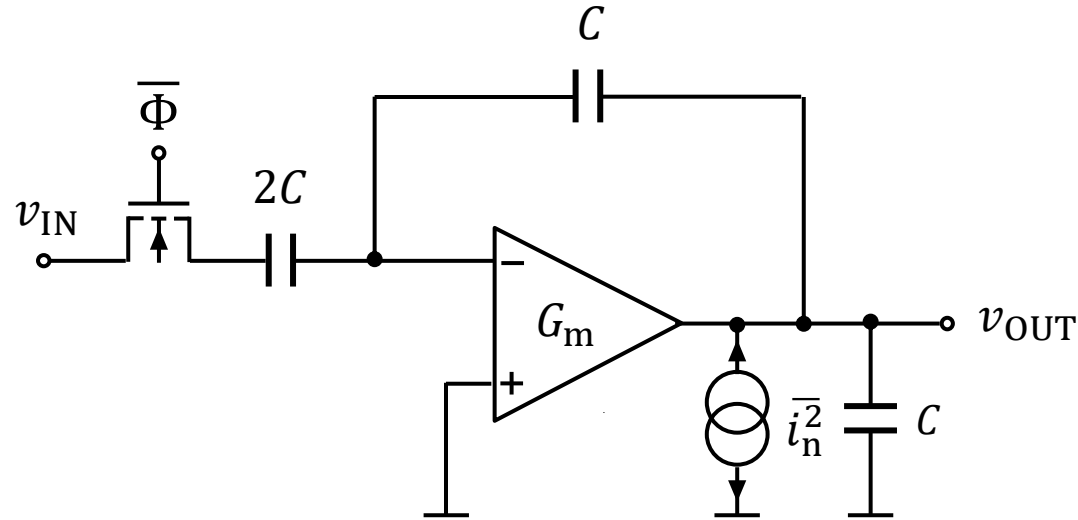


$$C_{L_{tot}} = C_L + (1 - \beta)C_F = \frac{5}{3}C$$

$$\overline{v_{n,out,R_{on}}^2} = PSD_{R_{on}} \cdot |STF|^2 \cdot NBW = 4kTR_{on} \cdot 4 \cdot \frac{1}{4} \frac{\beta G_m}{C_{L_{tot}}} \leq 100 \mu V^2$$

$$R_{on} \leq 3 \Omega$$

TI.c – Charge Injection

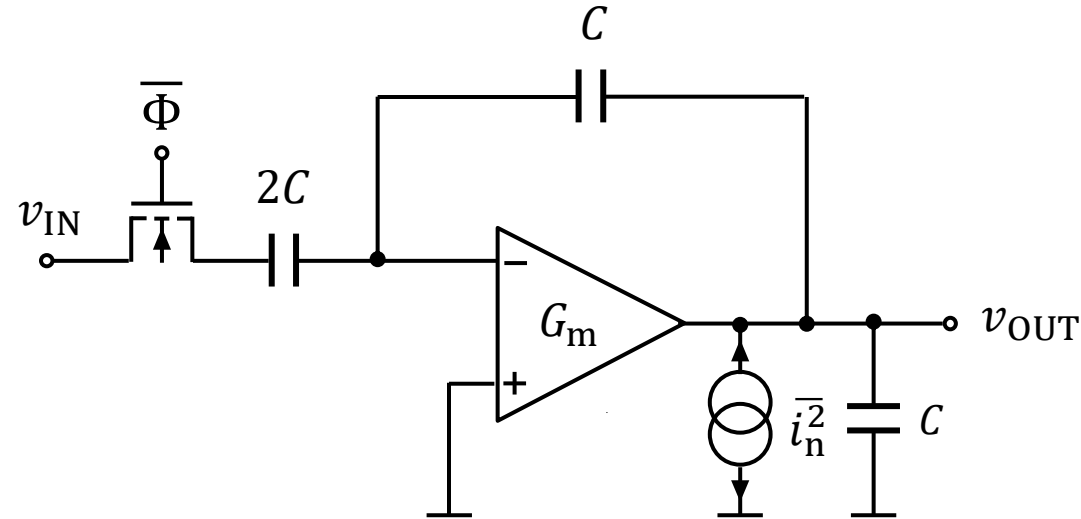


Assuming a 50/50 split of the channel charge results in

$$Q_{\text{ch}} = -\frac{1}{2} C_{\text{OV}} V_{\text{GS}_{\text{eff}}} = -\frac{1}{2} 50 \text{ fF} (V_{\text{GS}} - V_{\text{th}}) = -\frac{1}{2} 50 \text{ fF} (1.8 \text{ V} - 0.5 \text{ V} - 0.5 \text{ V}) = -0.02 \text{ pC}$$

on the left plate of $2C$.

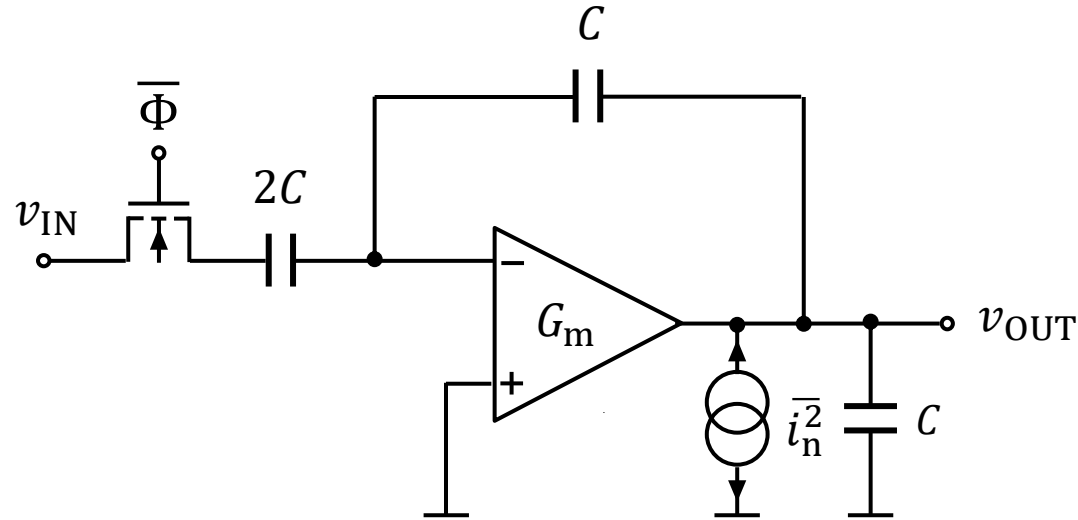
TI.c – Charge Injection



The charge on the right plate of the feedback capacitor C thus increases by 0.02 pC, whereby its voltage and thus the output voltage increases by

$$\Delta v_{\text{OUT, ch}} = \frac{0.02 \text{ pC}}{1 \text{ pF}} = 20 \text{ mV}$$

TI.c – Clock Feedthrough



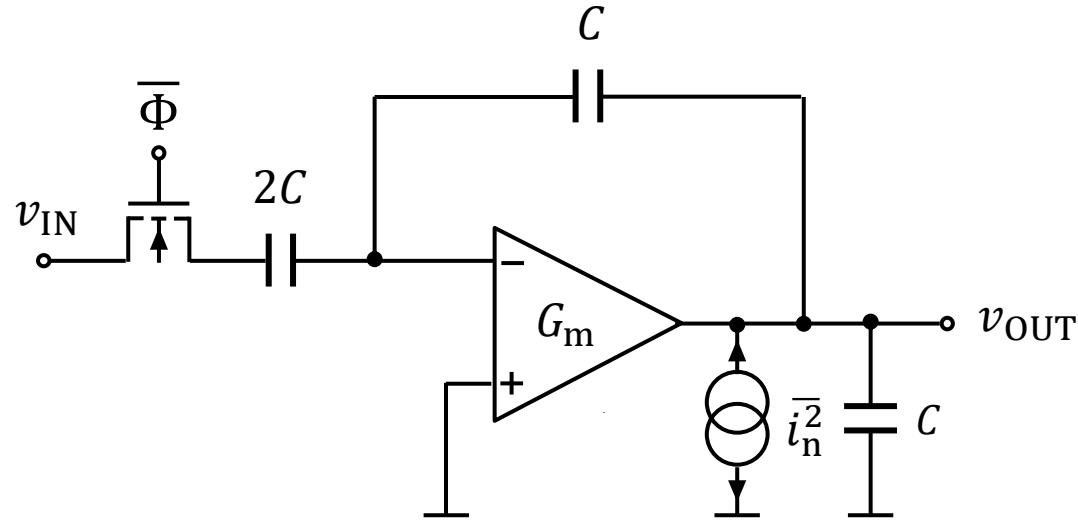
See lecture Chapter 5 – Sampling Circuits / Fast Gating:

The voltage across $C_S = 2C$ drops from v_{IN} to

$$v_{IN} - \Delta v_{CF} = v_{IN} - \frac{C_{OV}}{2C + C_{OV}} (\Phi_H - \Phi_L) = v_{IN} - \frac{10 \text{ fF}}{2 \text{ pF} + 10 \text{ fF}} 1.8 \text{ V} \approx v_{IN} - \mathbf{8.96 \text{ mV}}$$

whereby the (positive) charge on the left plate of $2C$ reduces by $-2 \text{ pF} \cdot 8.96 \text{ mV} = -17.92 \text{ fC}$

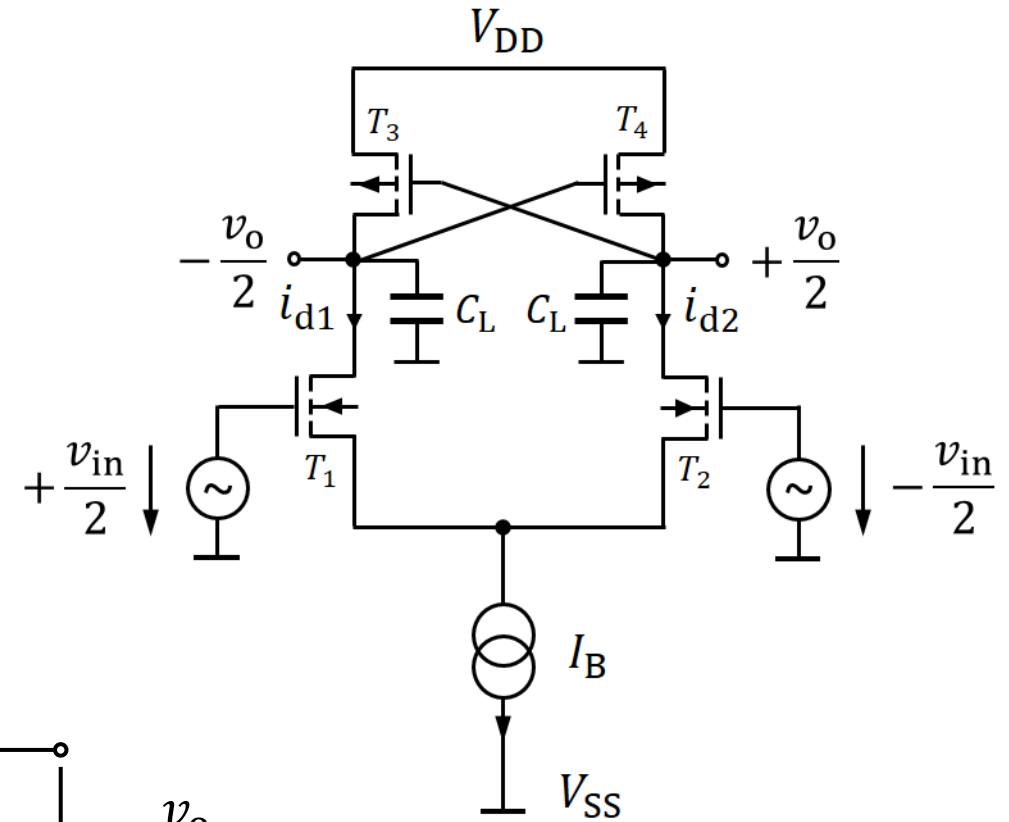
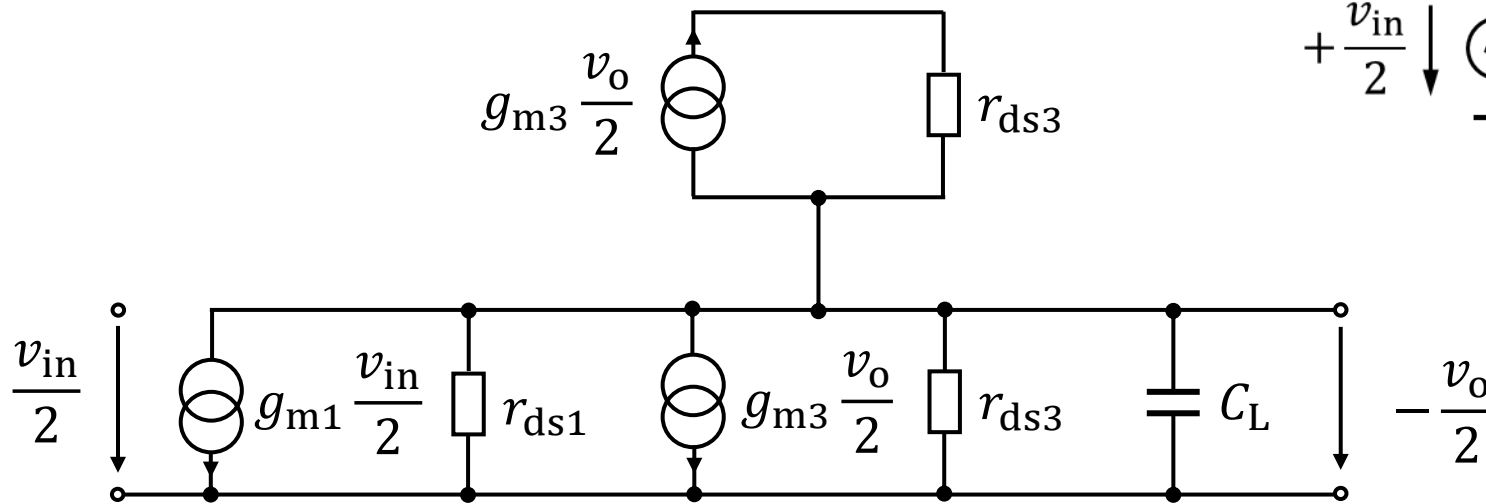
TI.c – Clock Feedthrough



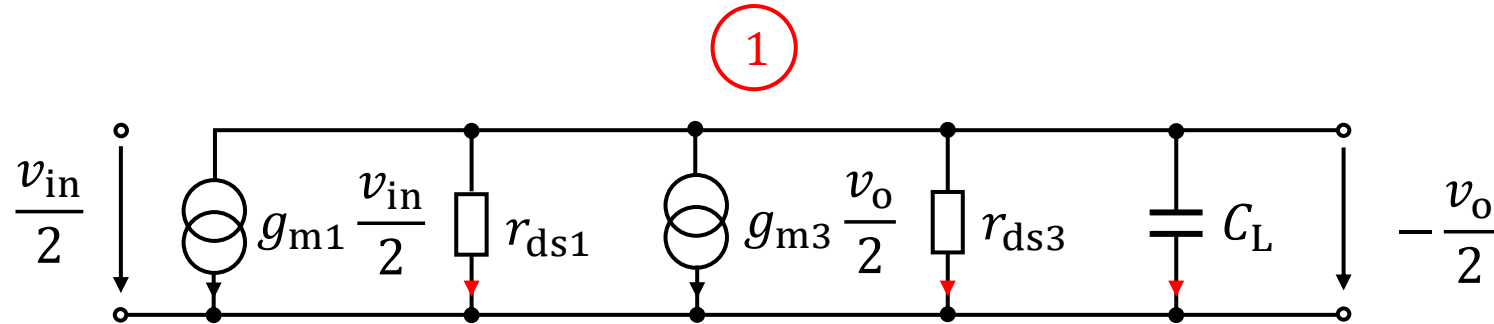
The charge on the right plate of the feedback capacitor C thus increases by 17.92 fC, whereby its voltage and thus the output voltage increases by

$$\Delta v_{\text{OUT,ci}} = \frac{17.92 \text{ fC}}{1 \text{ pF}} = 17.92 \text{ mV}$$

TII.a – Small-Signal Equivalent Circuit



TII.b – Step Response - Time-Domain Equation

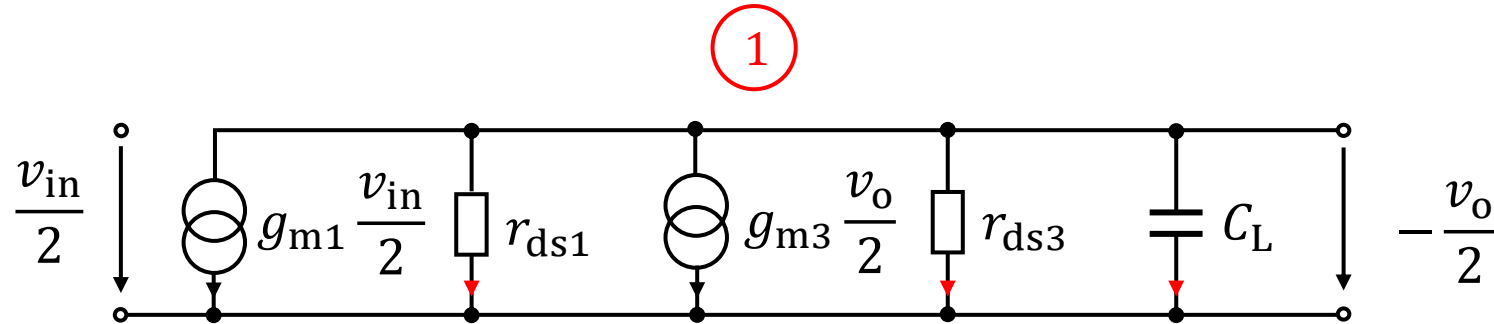


Application of Kirchhoff's Current Law to node 1:

$$g_{m1} \frac{v_{in}}{2} \frac{1}{s} - (g_{ds1} + g_{ds3} + sC_L) \frac{v_o(s)}{2} + g_{m3} \frac{v_o(s)}{2} = 0$$

$$\Leftrightarrow sC_L \frac{v_o(s)}{2} + (g_{m3} - g_{ds1} - g_{ds3}) \frac{v_o(s)}{2} = -g_{m1} \frac{v_{in}}{2} \frac{1}{s}$$

TII.b – Step Response - Time-Domain Equation

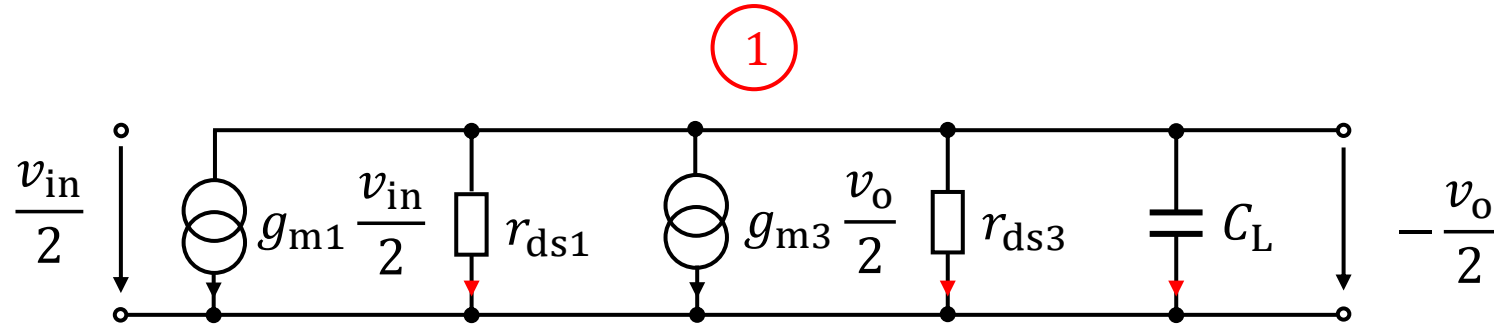


From frequency domain to time domain:

$$sC_L \frac{v_o(s)}{2} + (g_{m3} - g_{ds1} - g_{ds3}) \frac{v_o(s)}{2} = -g_{m1} \frac{v_{in}}{2} \frac{1}{s} \rightarrow C_L \frac{dv_o(t)}{dt} + (g_{m3} - g_{ds1} - g_{ds3}) v_o(t) = -g_{m1} v_{in}$$

$$\Leftrightarrow \frac{dv_o(t)}{dt} + \frac{(g_{m3} - g_{ds1} - g_{ds3})}{C_L} v_o(t) = -\frac{g_{m1}}{C_L} v_{in}$$

TII.b – Step Response - Time-Domain Equation



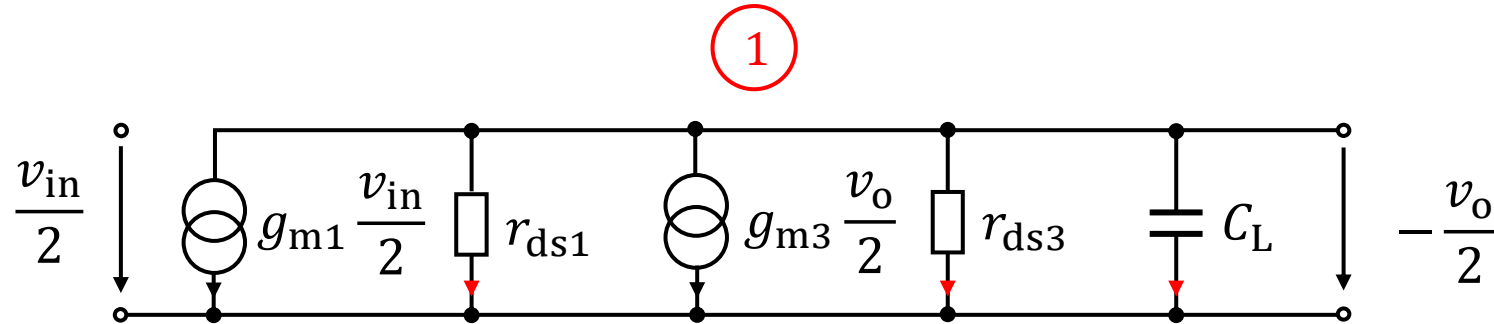
Comparison:

$$\frac{dv_o(t)}{dt} + \frac{g_{m3} - g_{ds1} - g_{ds3}}{C_L} v_o(t) = -\frac{g_{m1}}{C_L} v_{in} \Leftrightarrow \frac{dy(t)}{dt} + P \cdot y(t) = Q$$

$$\rightarrow P = \frac{g_{m3} - g_{ds1} - g_{ds3}}{C_L}$$

$$Q = -\frac{g_{m1}}{C_L} v_{in}$$

TII.b – Step Response - Time-Domain Equation



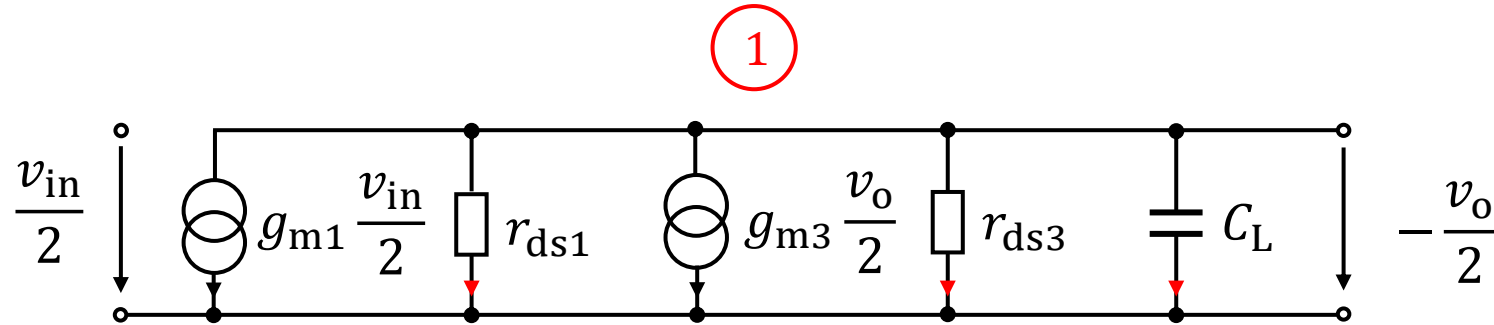
Since the solution of the ordinary differential equation is $y(t) \cdot e^{\int P dt} = \int Q \cdot e^{\int P dt} dt + C$, we get

$$y(t) \cdot e^{Pt} = \int Q \cdot e^{Pt} dt + C$$

$$\Leftrightarrow y(t) \cdot e^{Pt} = \frac{Q}{P} e^{Pt} + C$$

$$\Leftrightarrow y(t) = \frac{Q}{P} + C \cdot e^{-Pt}$$

TII.b – Step Response - Time-Domain Equation



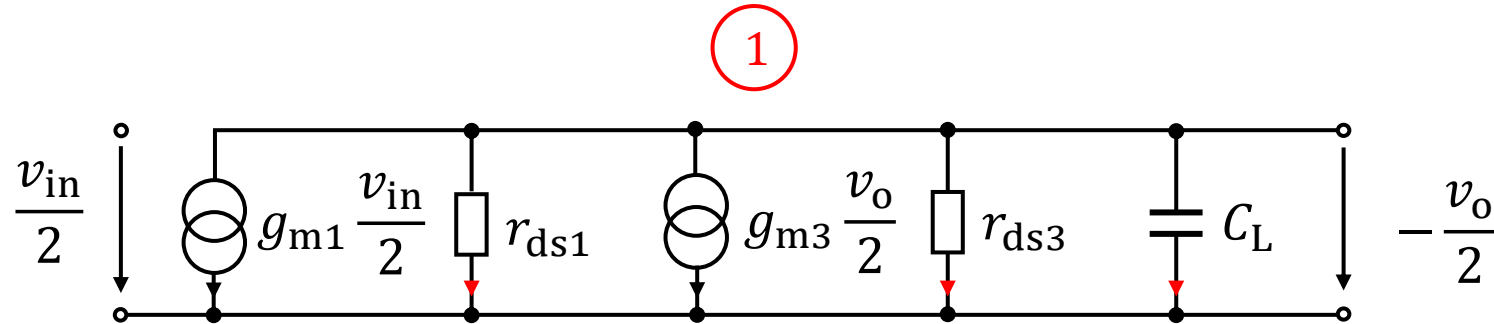
Since $y(t = 0 \text{ s}) = 0 \text{ V}$, C results in

$$y(t = 0 \text{ s}) = 0 \text{ V} = \frac{Q}{P} + C \cdot e^0 \rightarrow C = -\frac{Q}{P}$$

$$P = \frac{g_{m3} - g_{ds1} - g_{ds3}}{C_L}$$

$$Q = -\frac{g_{m1}}{C_L} v_{in}$$

TII.b – Step Response - Time-Domain Equation



Overall, the time-domain step response results in

$$v_o(t) = \frac{Q}{P} + C \cdot e^{-Pt} = \frac{Q}{P} (1 - e^{-Pt})$$

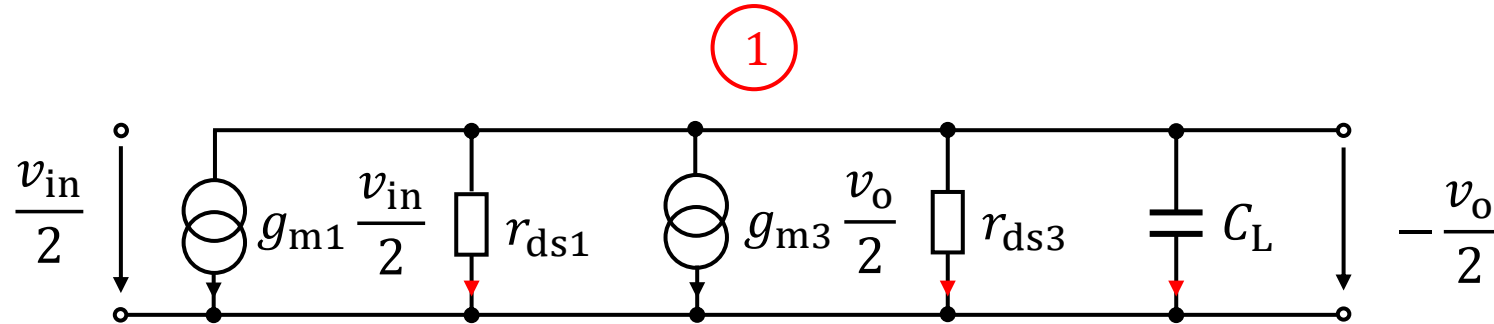
$$= \frac{g_{m1}}{g_{m3} - g_{ds1} - g_{ds3}} \left(1 - e^{-\frac{g_{m3} - g_{ds1} - g_{ds3}}{C_L} \cdot t} \right) v_{in}$$

$$C = -\frac{Q}{P}$$

$$P = \frac{g_{m3} - g_{ds1} - g_{ds3}}{C_L}$$

$$Q = -\frac{g_{m1}}{C_L} v_{in}$$

TII.b – Step Response - Time-Domain Equation



$$v_o(t) = \frac{g_{m1}}{g_{m3} - g_{ds1} - g_{ds3}} \left(1 - e^{-\frac{g_{m3} - g_{ds1} - g_{ds3}}{C_L} t} \right) v_{in}$$

$$= \frac{g_{m1}}{g_{m3} - g_{ds1} - g_{ds3}} \left(1 - e^{-t/\tau} \right) v_{in} \quad \text{with } \tau = \frac{C_L}{g_{m3} - g_{ds1} - g_{ds3}}$$

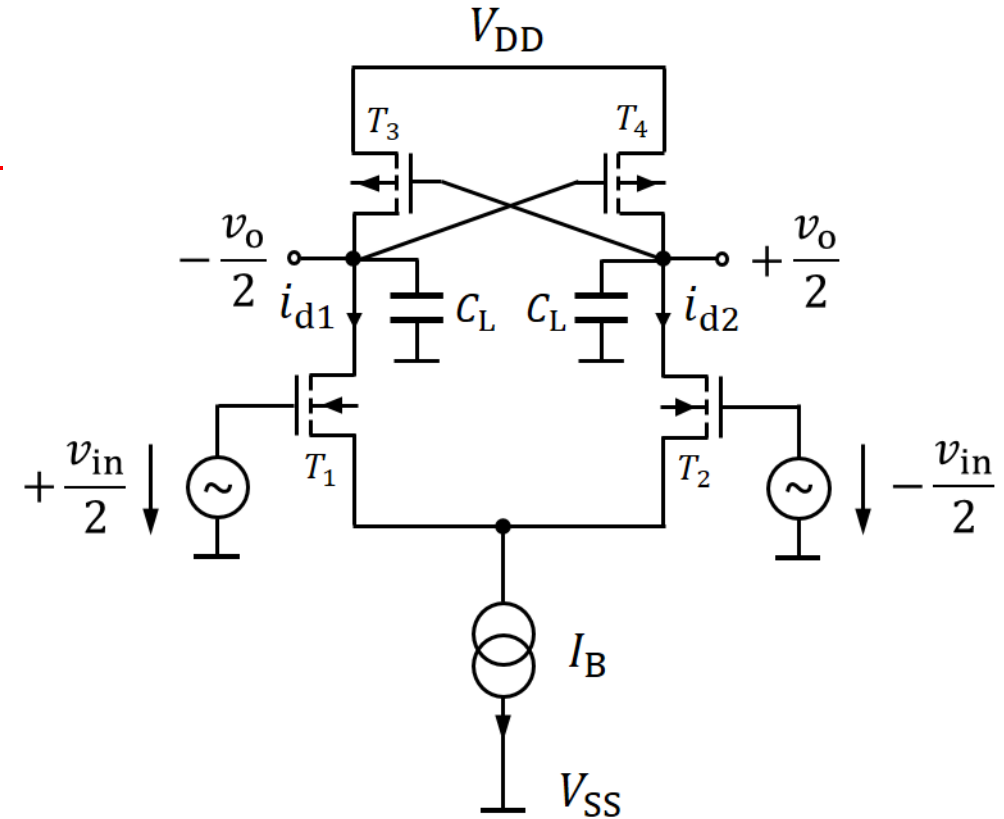
Considering the right-half circuit, we see that $v_o = V_{O,CM} + \frac{v_o}{2}$ determines the gate voltage V_G of transistor T_3 .

The transistor T_3 turns off if

$$|V_{GS3}| = V_{SG3} \leq |V_{th,p}|$$

$$\Leftrightarrow V_{DD} - \left(V_{CM,O} + \frac{v_o}{2}\right) \leq |V_{th,p}|$$

$$\Leftrightarrow v_o \geq 2(V_{DD} - V_{CM,O} - |V_{th,p}|)$$

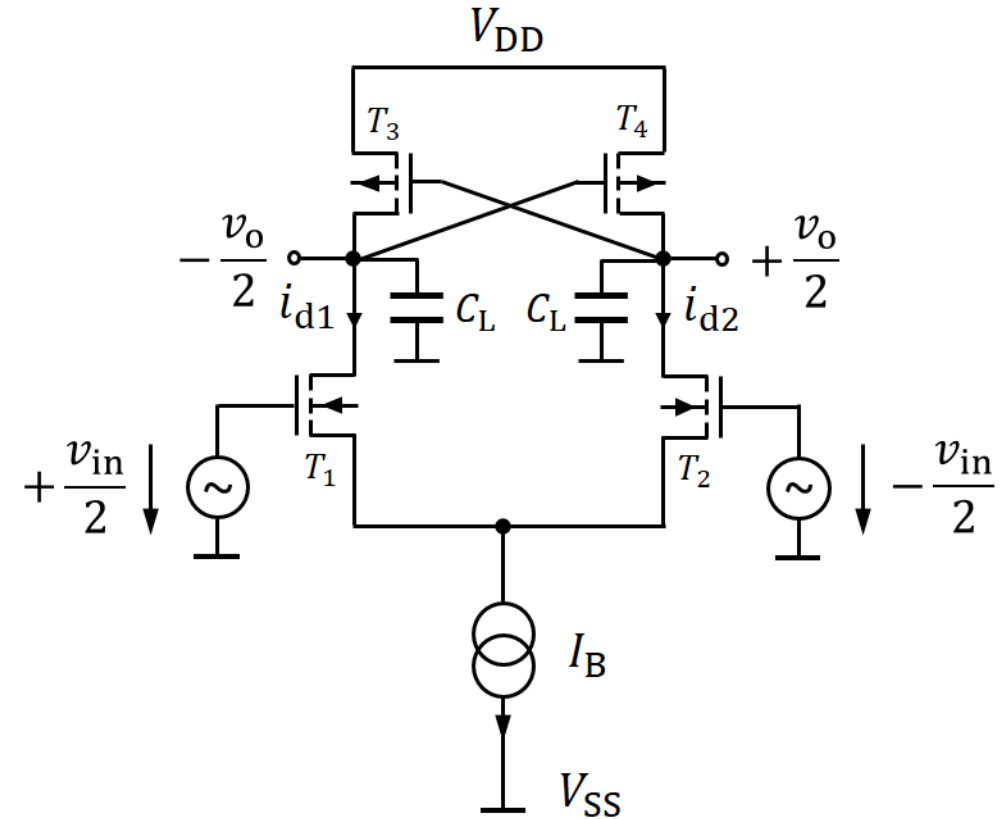


TII.d – Slewing Behavior

Since T_3 turns off for $v_o \geq 2(V_{DD} - V_{CM,O} - |V_{th,p}|)$,
the current $i_{D1} \approx I_B/2$ can be used
to charge or discharge the capacitor C_L .

The slew rate thus results in

$$SR = \frac{dv_o}{dt} = -\frac{I_B}{2C_L}$$



TII.e – Settling Time

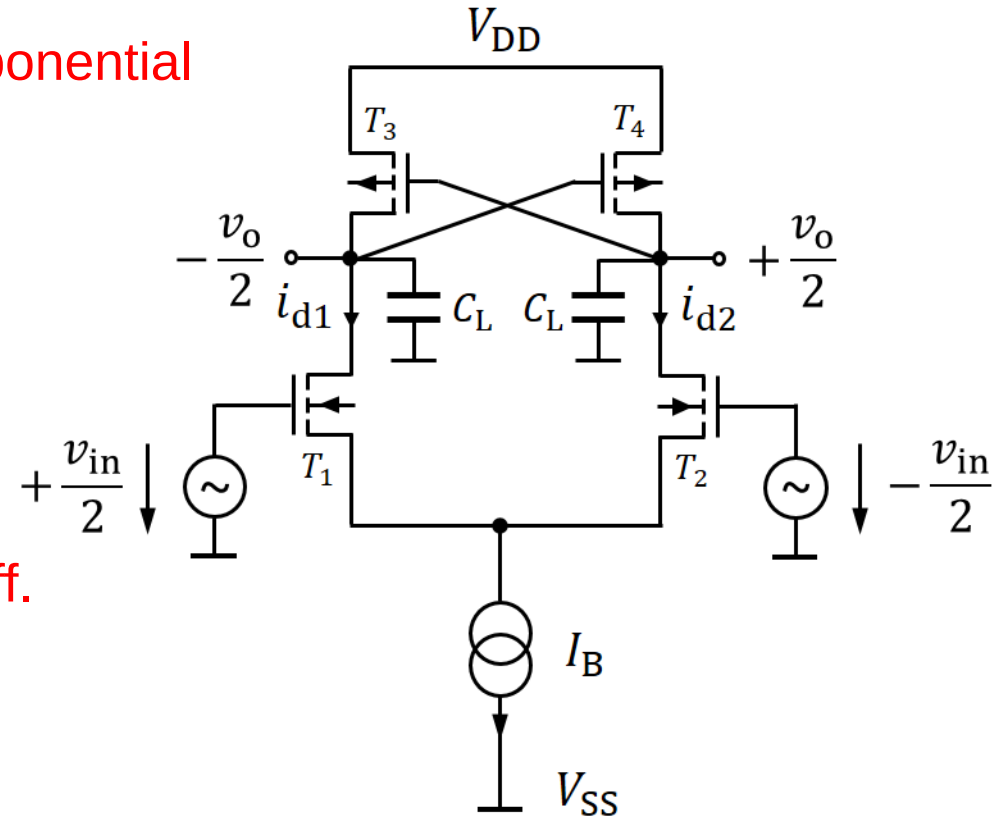
As derived in TII.c, the output voltages switches from exponential to linear settling if

$$v_o \geq 2(V_{DD} - V_{CM,O} - |V_{th,p}|)$$

$$\Leftrightarrow \frac{v_o}{2} \geq V_{DD} - V_{CM,O} - |V_{th,p}|$$

since once $v_o/2$ reached $V_{DD} - V_{CM,O} - |V_{th,p}|$, T_3 turns off.

$-v_o/2$ is then at $-V_{DD} + V_{CM,O} + |V_{th,p}|$ and, depending on the magnitude of v_{in} , continues dropping with a linear rate of $-I_B/(2C_L) \cdot t$

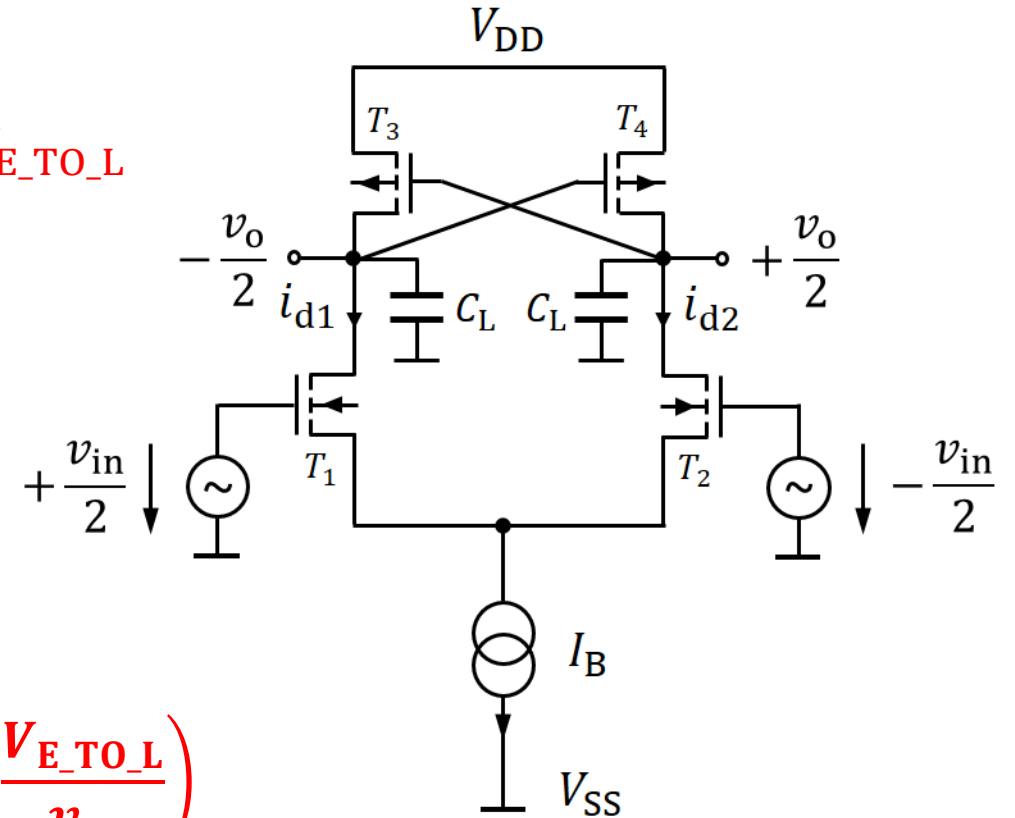


exponential settling for $0 \leq \frac{v_o}{2} \leq V_{DD} - V_{CM,O} - |V_{th,p}| = V_{E_TO_L}$

$$v_o(t) = \frac{g_{m1}}{g_{m3} - g_{ds1} - g_{ds3}} (1 - e^{-t/\tau}) v_{in}$$

with $\tau = \frac{C_L}{g_{m3} - g_{ds1} - g_{ds3}}$

$$t_{e_to_l} = -\frac{C_L}{g_{m3} - g_{ds1} - g_{ds3}} \log \left(1 - \frac{g_{m3} - g_{ds1} - g_{ds3}}{g_{m1}} \frac{V_{E_TO_L}}{v_{in}} \right)$$



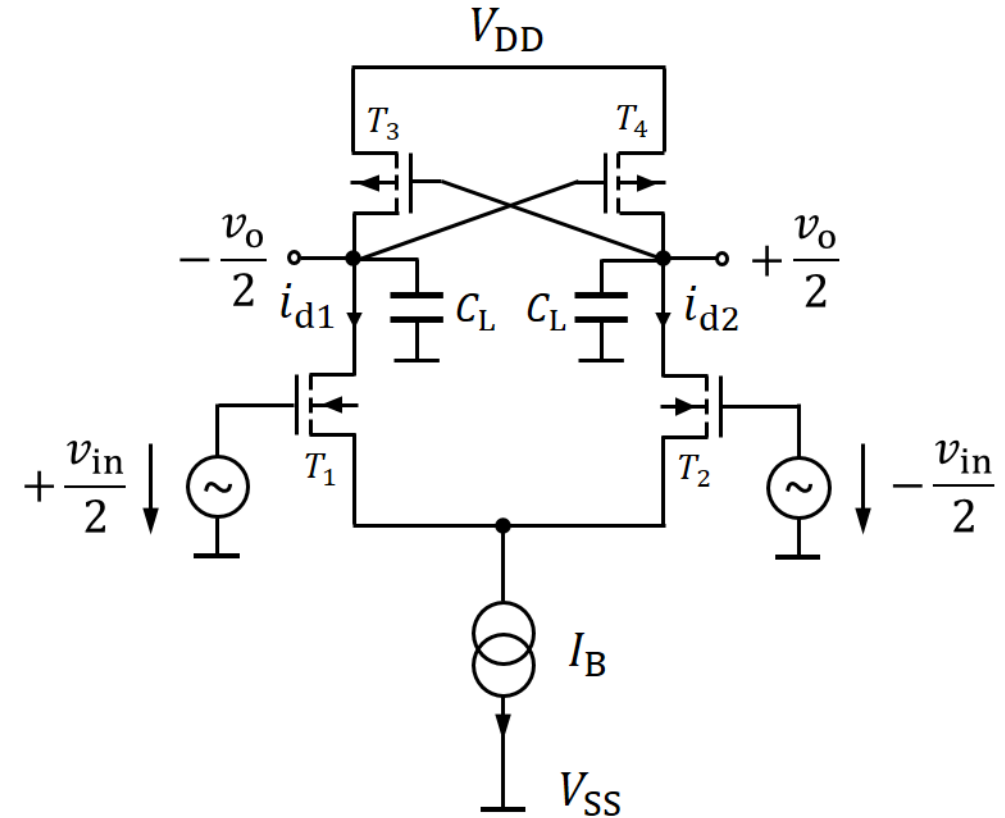
linear settling for the range

$$\Delta v_{\text{fall}} - (V_{\text{DD}} - V_{\text{CM},\text{O}} - |V_{\text{th},\text{p}}|) = \Delta v_{\text{fall}} - V_{\text{E_TO_L}}$$

$$v_{\text{o}}(t) = \frac{I_{\text{B}}}{2C_{\text{L}}} dt$$

$$\rightarrow v_{\text{o}}(t_1) = \Delta v_{\text{fall}} - V_{\text{E_to_L}} = \frac{I_{\text{B}}}{2C_{\text{L}}} t_1$$

$$\Leftrightarrow t_1 = \frac{2C_{\text{L}}}{I_{\text{B}}} \log(\Delta v_{\text{fall}} - V_{\text{E_TO_L}})$$



The total settling t_t time thus results in

$$t_t = t_{e_to_l} + t_l$$

$$= -\frac{C_L}{g_{m3} - g_{ds1} - g_{ds3}} \log \left(1 - \frac{g_{m3} - g_{ds1} - g_{ds3}}{g_{m1}} \frac{V_{E_TO_L}}{v_{in}} \right) + \frac{2C_L}{I_B} \log(\Delta v_{fall} - V_{E_to_L})$$

$$\text{with } V_{E_TO_L} = V_{DD} - V_{CM,O} - |V_{th,p}|$$

As derived in the lecture:

The DNL of code k is defined by the transitions T and the ideal code width w_{ideal} according to

$$DNL(k) = \frac{w(k) - w_{\text{ideal}}}{w_{\text{ideal}}} = \frac{T(k) - T(k-1) - w_{\text{ideal}}}{w_{\text{ideal}}}$$

with $DNL(k=1) = DNL(k=2^B) = \textit{undefined}$ since these codes have an infinite width.

Since the first and the last comparator are offset-free, we get

$$\sigma_{DNL(k=2)} = \sigma_{DNL(k=2^B-1)} = \sigma_{OS} = 2 \text{ mV} = 0.256 \text{ LSB}.$$

For all other codes, we get $\sigma_{DNL(3 \leq k \leq 2^B-2)} = \sqrt{2} \sigma_{OS} \approx 2.83 \text{ mV} \approx 0.362 \text{ LSB}.$

As derived in the lecture:

The INL of code k is defined by the transition T and the ideal code width w_{ideal} according to

$$INL(k) = \frac{T(k-1) - T_{\text{ideal}}(k-1)}{w_{\text{ideal}}}$$

with $INL(k=1) = \textit{undefined}$ since this code is “transition free”.

Since the first and the last comparator are offset-free, we get

$$\sigma_{\text{INL}}(k=2) = \sigma_{\text{INL}}(k=2^B) = 0.$$

For all other codes, we get $\sigma_{\text{INL}}(3 \leq k \leq 2^B-1) = \sigma_{\text{OS}} = 2 \text{ mV} \approx 0.256 \text{ LSB}$.